# Geometrical meaning of the Moore-Penrose pseudo inverse

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**Abstract**

A rectangular real valued linear system $Ax = y$ of size $m \times n$ is completely defined by a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$. Given $y \in \mathbb{R}^m$, by solving the system one attempts to find $x^* \in \mathbb{R}^n$ such that $y = Ax^*$. If $A$ is full column rank, its solution $x^*$ can be computed via the Moore-Penrose pseudo inverse $A^+$ as $x^* = A^+ y$. However, in general it is $Ax^* \neq y$. In this note we provide a geometrical interpretation of the Moore-Penrose pseudo inverse $A^+$ and we show that $Ax^*$ is the orthogonal projection of $y$ onto the subspace of $\mathbb{R}^m$ spanned by the columns of $A$.

## Preliminaries

A real valued matrix $A$ of size $m \times n$ can be seen as a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$ that maps $x \mapsto Ax$, where $Ax$ is the usual matrix-vector product. Two important spaces related to the linear map $A$ are:

- $\ker(A) \subset \mathbb{R}^n$ the kernel of $A$ is the vector subspace of $\mathbb{R}^n$ of all the points that are mapped to $0 \in \mathbb{R}^m$:

$$\ker(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

- $\mathrm{S}(A) \subset \mathbb{R}^m$ the space spanned by $A$ is the vector subspace of $\mathbb{R}^m$ of all the points that can be mapped by $A$:

$$\mathrm{S}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

The dimensions of these (vector) spaces depend on $k = \mathrm{rank}(A)$ (the rank of $A$), which is the the maximum number of linearly independent rows/columns of matrix $A$. The *Dimension Theorem* states that

$$\dim(\ker(A)) = n - k, \qquad \dim(\mathrm{S}(A)) = k. \tag{1}$$

Notice that, if $k = \min\{n, m\}$, then matrix $A$ has full rank and we have $\dim(\ker(A)) = 0$, so $\ker(A) = \{0\}$. In order to describe the geometrical meaning of the Moore-Penrose pseudo inverse, we need to first introduce the notion of orthogonal complement.

**Definition** (Orthogonal complement)**.** *Let $E, V$ be two linear spaces with $E \subset V$ subspace. We define $E^\perp$ the orthogonal complement of $E$ in $V$ to be*

$$E^\perp = \{y \in V : y^T z = 0 \text{ for all } z \in E\}.$$

*It can be easily proved that $E^\perp \subset V$ is linear subspace. Moreover, if $E \subset V$ is closed subspace, then $E$ and its orthogonal complement are in direct sum and generate $V$:*

$$V = E \oplus E^\perp, \qquad E \cap E^\perp = 0.$$

*As important consequence of this, we have that every $y \in V$ can be* ***uniquely decomposed*** *into the sum of two orthogonal components:*

$$y = q + z \qquad q \in E,\, z \in E^\perp.$$

*To emphasize the fact that $q, z$ are unique, they depend on $y$ and $E$ and they lie on orthogonal spaces, we better write $P_E(y) := q$ and $P_{E^\perp}(Y) := z$. Thus*

$$y = P_E(y) + P_{E^\perp}(y) \qquad P_E(y) \in E,\, P_{E^\perp}(y) \in E^\perp.$$

*Operator $P_X(.)$ is sometimes called orthogonal projection onto space $X$.*

## The Moore-Penrose pseudo inverse

Let us consider a general $y \in \mathbb{R}^m$. Of course, if $y \in \mathbb{R}^m \setminus \mathrm{S}(A)$ the linear system $y = Ax$ has no solution. Whereas, if $y \in \mathrm{S}(A)$ the lineary system $y = Ax$ has at least one solution.

Being $\mathrm{S}(A) \subset \mathbb{R}^m$ a finite dimensional space, it is also closed linear subspace. Therefore the orthogonal complement decomposition given above applies and we have:

$$\mathbb{R}^m = \mathrm{S}(A) \oplus \mathrm{S}(A)^\perp.$$

This implies mainly two things:

- $\mathbb{R}^m \setminus \mathrm{S}(A) = \mathrm{S}(A)^\perp$, therefore the $y$ for which the linear system $y = Ax$ cannot be solved lie in the orthogonal space of $\mathrm{S}(A)$, i.e., $y \in \mathrm{S}(A)^\perp$.
- We can write every $y \in \mathbb{R}^m$ uniquely as a sum of two orthogonal components as

  $$y = P_{\mathrm{S}(A)}(y) + P_{\mathrm{S}(A)^\perp}(y) = q + z.$$

  where $q \in \mathrm{S}(A)$ and $z \in \mathrm{S}(A)^\perp$ are used in the following to simplify the notation.

By using the definition of orthogonal component and the definition of $\mathrm{S}(A)$, we can now see that

$$\begin{aligned}
\mathrm{S}(A)^\perp &= \{b \in \mathbb{R}^m : b^T a = 0 \text{ for all } a \in \mathrm{S}(A)\} \\
&= \{b \in \mathbb{R}^m : b^T Ax = 0 \text{ for all } x \in \mathbb{R}^n\} \\
&= \{b \in \mathbb{R}^m : x^T A^T b = 0 \text{ for all } x \in \mathbb{R}^n\} \\
&= \{b \in \mathbb{R}^m : A^T b = 0\} \\
&= \ker(A^T)
\end{aligned}$$

where $A^T : \mathbb{R}^m \to \mathbb{R}^n$ is the linear map obtained by transposing matrix $A$. By applying $A^T$ to our $y \in \mathbb{R}^m$ and using the definition of kernel, we get

$$A^T y = A^T(q + z) = A^T q + A^T z = A^T q + 0 = A^T q.$$

where $A^T z = 0$ holds because $z \in \mathrm{S}(A)^\perp = \ker(A^T)$. Since $q \in \mathrm{S}(A)$ there exists $x \in \mathbb{R}^n$ such that $Ax = q$, thus

$$A^T y = A^T A x. \tag{2}$$

In order to *constructively* find such $x$, we need to inspect $k = \mathrm{rank}(A)$.

- **Full row rank.** If $m \leq n$ and $A$ is full rank (i.e., $k = m$) we say that $A$ is *full row rank*. In this case (1) implies $\dim(\mathrm{S}(A)) = m$, therefore the linear system $y = Ax$ admits at least one solution. In the particular case $n = m$, the solution is $y = A^{-1}x$.

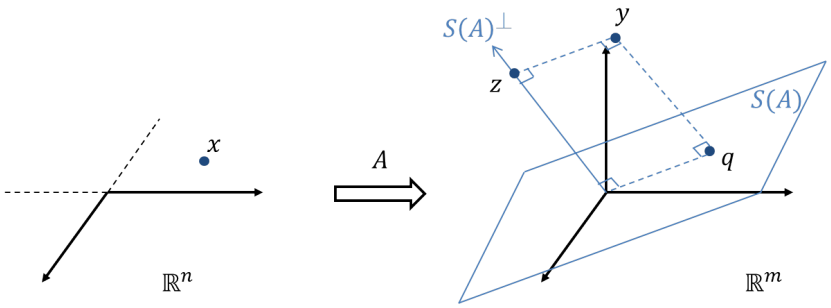
- **Full column rank.** If $m \geq n$ and $A$ is full rank (i.e., $k = n$) we say that $A$ is *full column rank*. In this case $A^T A$ is invertible, thus from (2) we get

$$x = (A^T A)^{-1} A^T y. \tag{3}$$

  and such $x$ is unique. Notice that $Ax = q = P_{\mathrm{S}(A)}(y)$, so the solution of the linear system $y = Ax$ provided by (3) is in general the *orthogonal projection of $y$ onto the space spanned by $A$*. The matrix $A^+ := (A^T A)^{-1} A^T$ is called the Moore-Penrose pseudo inverse of $A$. It is called as such because in the particular case $A$ is invertible (i.e., $m = n = k$), then $A^+ = A^{-1}$, in fact

$$A^+ = (A^T A)^{-1} A^T = A^{-1} (A^T)^{-1} A^T = A^{-1}.$$

  The geometrical meaning of $A^+$ is illustrated in the figure below. Given $y \in \mathbb{R}^m$, then $x = A^+ y \in \mathbb{R}^n$ is the unique point such that $Ax = q$ is the orthogonal projection of $y$ onto $\mathrm{S}(A)$.




## Numerical example

We develop here a simple test case that shows the geometrical meaning of the Moore-Penrose inverse. We consider the linear map $A : \mathbb{R}^2 \to \mathbb{R}^3$ and the point $y$ given by

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Matrix $A$ maps $\mathbb{R}^2$ into the 2-dimensional subspace of $\mathbb{R}^3$ given by:

$$\mathrm{S}(A) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_3 = 0\}$$

and $y$ does not belong to such space. The orthogonal projection of $y$ onto $\mathrm{S}(A)$ is precisely $q = [1, 1, 0]^T$, which is obtained by multiplying $A$ with $x = [1, 1]^T$. So, we expect the result of $A^+ y$ to be exactly $x$. There it is:

```
%A maps R^2 into the subspace of R^3 given by (z=0)
A=[1,0;0,1;0,0];
%y is the point of coordinates (1,1,1)
y=[1;1;1];
%apply pseudo inverse
x=pinv(A)*y
>> x =
        1
        1
```